Deep Learning Research Report · Rohail Ahmad · 2024 rohailahmad-code/Lung-Cancer-Histopathological-Images-Classification

Deep Learning-Based Lung Cancer Classification using Histopathological Images

Rohail Ahmad

AI & Machine Learning Engineer · Computer Vision Research

Technical Report, 2024

github.com/rohailahmad-code/Lung-Cancer-Histopathological-Images-Classification

ABSTRACT

Lung cancer remains one of the deadliest malignancies worldwide, with early and accurate classification being paramount to patient survival. This report presents a comparative deep learning study evaluating two distinct architectures — **AlexNet** and **YOLOv9** — for automated multi-class classification of lung cancer subtypes from histopathological tissue images. The LC25000 dataset, comprising 25,000 color images across three lung tissue classes (Adenocarcinoma, Squamous Cell Carcinoma, and Benign), was used for training, validation, and testing. AlexNet was evaluated in its classical CNN classification role, while YOLOv9 was adapted for classification via its object detection backbone. Performance was benchmarked across accuracy, precision, recall, F1-score, and inference latency. Our analysis provides actionable insights into the trade-offs between architectural complexity, detection speed, and diagnostic accuracy in a clinical-grade pathology context.

Keywords: Lung Cancer, Histopathological Image Classification, Deep Learning, AlexNet, YOLOv9, Convolutional Neural Networks, Medical Imaging, LC25000.

1. Introduction

Lung cancer is the leading cause of cancer-related mortality globally, accounting for approximately 18% of all cancer deaths annually (World Health Organization, 2023). The two most prevalent subtypes — Adenocarcinoma and Squamous Cell Carcinoma — together constitute over 80% of non-small cell lung cancer (NSCLC) diagnoses. Histopathological analysis through microscopic examination of tissue samples remains the gold standard for definitive diagnosis; however, this process is time-consuming, costly, and subject to inter-observer variability among pathologists.

The emergence of deep learning has demonstrated transformative potential in computational pathology. Convolutional Neural Networks (CNNs) have proven capable of learning discriminative visual features from tissue images that correlate strongly with expert diagnoses. This project investigates whether modern architectures — specifically

AlexNet and the recently released YOLOv9 — can achieve clinically relevant performance in distinguishing between lung cancer subtypes and benign tissue.

1.1 Motivation

The primary motivation is to build an automated, scalable, and accurate diagnostic assistant that can reduce diagnostic time and support pathologists in high-throughput screening environments. Comparing classical CNN designs against state-of-the-art detection-adapted architectures offers insight into which approaches best suit the computational pathology domain.

1.2 Objectives

- Implement AlexNet for multi-class lung cancer classification from histopathological images.
- Adapt YOLOv9 for classification using its feature extraction backbone.
- Perform a rigorous comparative analysis across standard evaluation metrics.
- Identify strengths, limitations, and optimal use cases for each architecture in pathology settings.
- Discuss clinical deployment considerations and future research directions.

2. Related Work

Deep learning applications in histopathological image analysis have grown rapidly since the seminal work of Krizhevsky et al. (2012), which introduced AlexNet and demonstrated the superiority of deep CNNs on large-scale visual recognition tasks. Subsequent architectures including VGGNet, ResNet, and DenseNet have been applied to various cancer classification tasks with increasing success.

In lung cancer specifically, Borkowski et al. (2019) introduced the LC25000 dataset — a balanced, high-quality collection of 25,000 histopathological images of lung and colon tissue — which has since become a benchmark for deep learning classification research. Studies using ResNet-50 on this dataset have reported accuracies exceeding 99%, while lighter architectures like MobileNet achieved 97–98% with significantly faster inference times.

The use of YOLO-family models for medical image analysis is comparatively newer. YOLOv5 was successfully employed for cell detection and mitosis counting in breast cancer pathology slides (Mahmood et al., 2022). YOLOv9, released by Wang et al. in 2024, introduced Programmable Gradient Information (PGI) and Generalized Efficient Layer Aggregation Network (GELAN) to improve both accuracy and efficiency, making it a compelling candidate for adaptation to classification tasks.

AlexNet YOLOv9 LC25000 Dataset ResNet Transfer Learning Computational Pathology

3. Dataset

3.1 LC25000 Histopathological Image Dataset

The LC25000 dataset contains **25,000 color images** at 768×768 pixel resolution, augmented to 5,000 samples per class from 750 unique tissue images. For this project, only the **lung tissue subset** was utilized, comprising three classes:

Table 1: Dataset Class Distribution (Lung Tissue Subset)

Class	Label	Samples	Description
Lung Adenocarcinoma	lung_aca	5,000	Malignant glandular tissue
Lung Squamous Cell Carcinoma	lung_scc	5,000	Malignant squamous tissue
Lung Benign Tissue	lung_n	5,000	Normal/healthy lung tissue

3.2 Data Preprocessing

- **Resizing:** All images resized to 227×227 for AlexNet and 640×640 for YOLOv9.
- **Normalization:** Pixel values normalized to [0, 1] using ImageNet mean and standard deviation.
- **Augmentation:** Random horizontal/vertical flipping, rotation (±15°), brightness and contrast jitter applied to the training set.
- **Split Ratio:** 70% training / 15% validation / 15% test (stratified by class).

[Representative Histopathological Images]
Left: Adenocarcinoma (glandular pattern) · Center: Squamous Cell Carcinoma (keratin pearls) · Right: Benign tissue

Figure 1: Sample images from the LC25000 lung tissue subset.

4. Methodology

4.1 AlexNet Architecture

AlexNet, introduced by Krizhevsky et al. (2012) and winner of the ImageNet LSVRC-2012 challenge, was one of the first deep CNNs to demonstrate the power of GPU-accelerated deep learning. The architecture consists of five convolutional layers followed by three fully connected layers, totaling approximately 60 million parameters.

AlexNet Configuration

- **Input:** 227×227×3 RGB images
- **Conv1:** 96 filters, 11×11, stride 4 + ReLU + MaxPool
- **Conv2:** 256 filters, 5×5, padding 2 + ReLU + MaxPool
- **Conv3:** 384 filters, 3×3, padding 1 + ReLU
- **Conv4:** 384 filters, 3×3, padding 1 + ReLU

YOLOv9 Configuration

- **Input:** 640×640×3 RGB images
- **Backbone:** GELAN (Generalized ELAN)
- **Neck:** PGI (Programmable Gradient Information)
- **Head:** Modified for 3-class classification
- **Variant:** YOLOv9-C (medium scale)
- **Pre-training:** COCO-pretrained weights (fine-tuned)

- **Conv5:** 256 filters, 3×3, padding 1 + ReLU + MaxPool
- **FC6:** 4096 units + ReLU + Dropout(0.5)
- **FC7:** 4096 units + ReLU + Dropout(0.5)
- **FC8:** 3 units (Softmax output)
- **Parameters:** ~60M

- **Feature pooling:** Global Average Pooling applied to detection head
- **Output:** 3-class Softmax classification
- **Parameters:** ~51M

4.2 Training Configuration

Table 2: Hyperparameter Configuration

Hyperparameter	AlexNet	YOLOv9
Optimizer	SGD (momentum=0.9)	Adam (β_1=0.9, β_2=0.999)
Learning Rate	0.001 (step decay)	0.001 (cosine annealing)
Batch Size	64	16
Epochs	50	50
Weight Decay	0.0005	0.0005
Dropout Rate	0.5 (FC layers)	0.1 (head)
Loss Function	Cross-Entropy	Cross-Entropy
Hardware	NVIDIA GPU	NVIDIA GPU
Framework	PyTorch	PyTorch / Ultralytics

4.3 Evaluation Metrics

- **Accuracy:** Overall proportion of correctly classified samples.
- **Precision (macro):** Average ratio of true positives to predicted positives across classes.
- **Recall (macro):** Average ratio of true positives to actual positives across classes.
- **F1-Score (macro):** Harmonic mean of precision and recall.
- **Inference Latency:** Average time per image on CPU (milliseconds).

5. Experimental Results

5.1 Quantitative Performance

98.4%	99.1%	12ms	28ms
AlexNet Accuracy	YOLOv9 Accuracy	AlexNet Latency	YOLOv9 Latency

Table 3: Comparative Performance on Test Set

Model	Accuracy	Precision	Recall	F1-Score	Latency (ms/img)
AlexNet (from scratch)	98.4%	98.2%	98.5%	98.3%	**12**
YOLOv9-C (fine-tuned)	**99.1%**	**99.0%**	**99.2%**	**99.1%**	28

5.2 Per-Class Results

Table 4: Per-Class Precision, Recall, and F1-Score (AlexNet)

Class	Precision	Recall	F1-Score	Support
Lung Adenocarcinoma	97.8%	98.9%	98.3%	750
Squamous Cell Carcinoma	98.5%	98.1%	98.3%	750
Benign Tissue	98.3%	98.4%	98.4%	750
Macro Avg	**98.2%**	**98.5%**	**98.3%**	2250

Table 5: Per-Class Precision, Recall, and F1-Score (YOLOv9-C)

Class	Precision	Recall	F1-Score	Support
Lung Adenocarcinoma	98.9%	99.3%	99.1%	750
Squamous Cell Carcinoma	99.2%	99.1%	99.2%	750
Benign Tissue	98.9%	99.2%	99.0%	750
Macro Avg	**99.0%**	**99.2%**	**99.1%**	2250

5.3 Training Convergence

AlexNet converged within approximately 35 epochs, demonstrating stable and consistent loss reduction. YOLOv9 benefited from its COCO-pretrained initialization, achieving near-peak validation accuracy by epoch 20 with cosine annealing. Both models showed no significant overfitting.

5.4 Confusion Matrix Analysis

The most common misclassifications in both models occurred between Adenocarcinoma and Squamous Cell Carcinoma — the two malignant classes. This is clinically expected, as borderline cases can exhibit overlapping histological features. Benign tissue was classified with near-perfect accuracy by both architectures.

6. Discussion

6.1 Architecture Trade-offs

Table 6: Architecture Trade-off Summary

Dimension	AlexNet	YOLOv9
Accuracy	98.4%	99.1% ✓
Inference Speed	12ms ✓	28ms
Training Ease	Straightforward ✓	Requires adaptation
Transfer Learning	Optional	COCO pretrained ✓
Clinical Deployment	Edge-friendly ✓	Server-recommended

6.2 Clinical Implications

- **Domain Shift:** Models may not generalize to images from different scanners or staining protocols.
- **Class Imbalance:** Real-world datasets frequently exhibit imbalanced class distributions.
- **Interpretability:** Pathologists require explainable predictions; Grad-CAM visualizations would improve trust.
- **Regulatory Compliance:** FDA and CE mark requirements necessitate prospective clinical validation.

6.3 Limitations

- Experiments conducted on a single, augmentation-heavy benchmark dataset.
- YOLOv9 was not originally designed for classification tasks; adaptation introduces architectural compromises.
- Computational requirements for YOLOv9 may limit deployment in resource-constrained settings.

7. Future Work

- **Multi-dataset generalization:** Evaluate on TCGA lung cancer slides and external hospital datasets.
- **Attention mechanisms:** Incorporate Vision Transformers (ViT) for comparison against CNN baselines.
- **Grad-CAM explanations:** Generate class activation maps to highlight discriminative tissue regions.

- **Whole Slide Image (WSI) analysis:** Extend the pipeline to handle gigapixel WSIs using patch-based approaches.
- **Clinical trial integration:** Prospective validation with board-certified pathologist review.

8. Conclusion

> This study presented a rigorous comparative analysis of AlexNet and YOLOv9 for lung cancer histopathological image classification. YOLOv9 achieved superior accuracy (99.1%) and F1-score, benefiting from its advanced GELAN backbone and COCO pretrained weights. AlexNet, while slightly less accurate (98.4%), demonstrated significantly faster inference (12ms vs. 28ms per image), making it better suited for edge or real-time screening applications. Both models clearly demonstrated that deep learning is a viable and powerful approach for automated lung cancer pathology, with the potential to substantially assist pathologists in clinical workflows.

9. References

1. Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). *ImageNet Classification with Deep Convolutional Neural Networks.* NIPS, 25.
2. Borkowski, A. A., et al. (2019). *Lung and Colon Cancer Histopathological Image Dataset (LC25000).* arXiv:1912.12142.
3. Wang, C.-Y., Yeh, I.-H., & Liao, H.-Y. M. (2024). *YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information.* arXiv:2402.13616.
4. Mahmood, T., et al. (2020). *Artificial Intelligence-Based Mitosis Detection in Breast Cancer Histopathology Images.* Journal of Clinical Medicine, 9(3), 749.
5. He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition.* CVPR 2016.
6. Litjens, G., et al. (2017). *A survey on deep learning in medical image analysis.* Medical Image Analysis, 42, 60–88.
7. World Health Organization. (2023). *Cancer Fact Sheet: Lung Cancer.* WHO Global Cancer Observatory.
8. Esteva, A., et al. (2017). *Dermatologist-level classification of skin cancer with deep neural networks.* Nature, 542(7639), 115–118.

© 2024 Rohail Ahmad · Technical Report · All rights reserved · GitHub Repository